(c) The disk $x^2 + y^2 < 1$ is open and bounded, but is not closed.
(d) The half-plane $y > 0$ is open, but is neither closed nor bounded.

The reason that closed and bounded regions are useful is the following theorem, which is also true for functions of three or more variables:[5]

Theorem 15.1: Extreme Value Theorem for Multivariable Functions

If f is a continuous function on a closed and bounded region R, then f has a global maximum at some point (x_0, y_0) in R and a global minimum at some point (x_1, y_1) in R.

If f is not continuous or the region R is not closed and bounded, there is no guarantee that f achieves a global maximum or global minimum on R. In Example 4, the function g is continuous but does not achieve a global maximum or minimum in 2-space, a region which is closed but not bounded. Example 6 illustrates what can go wrong when the region is bounded but not closed.

Example 6 Does the function f have a global maximum or minimum on the region R given by $0 < x^2 + y^2 \leq 1$?

$$f(x, y) = \frac{1}{x^2 + y^2}$$

Solution The region R is bounded, but it is not closed since it does not contain the boundary point $(0, 0)$. We see from the graph of $z = f(x, y)$ in Figure 15.25 that f has a global minimum on the circle $x^2 + y^2 = 1$. However, $f(x, y) \to \infty$ as $(x, y) \to (0, 0)$, so f has no global maximum.

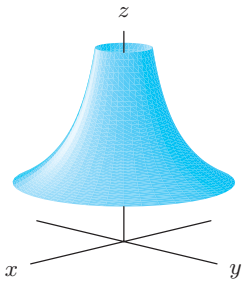


Figure 15.25: Graph showing $f(x, y) = \frac{1}{x^2+y^2}$ has no global maximum on $0 < x^2 + y^2 \leq 1$

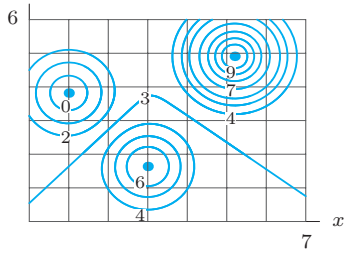
Exercises and Problems for Section 15.2

Exercises

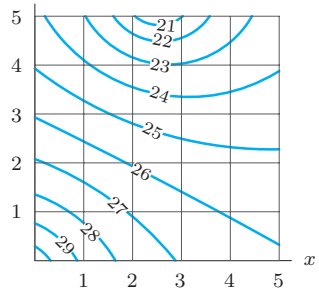
1. By looking at the weather map in Figure 12.1 on page 666, find the maximum and minimum daily high temperatures in the states of Mississippi, Alabama, Pennsylvania, New York, California, Arizona, and Massachusetts.

In Exercises 2–4, estimate the position and approximate value of the global maxima and minima on the region shown.

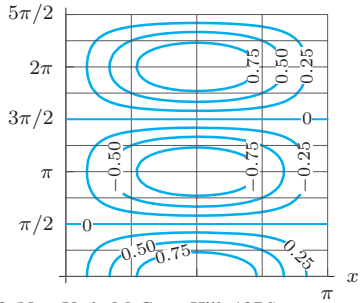
2.



3.



4.



[5]For a proof, see W. Rudin, *Principles of Mathematical Analysis*, 2nd ed., p. 89 (New York: McGraw-Hill, 1976).

In Exercises 5–7, find the global maximum and minimum of the function on $-1 \le x \le 1$, $-1 \le y \le 1$, and say whether it occurs on the boundary of the square. [Hint: Use graphs.]

5. $z = x^2 + y^2$ **6.** $z = -x^2 - y^2$ **7.** $z = x^2 - y^2$

In Exercises 8–12, do the functions have global maxima and minima in the xy-plane?

8. $f(x, y) = x^2 - 2y^2$ **9.** $g(x, y) = x^2y^2$

10. $h(x, y) = x^3 + y^3$ **11.** $f(x, y) = -2x^2 - 7y^2$

12. $f(x, y) = x^2/2 + 3y^3 + 9y^2 - 3x$

Problems

13. **(a)** Compute and classify the critical points of $f(x, y) = 2x^2 - 3xy + 8y^2 + x - y$.
(b) By completing the square, plot the contour diagram of f and show that the local extremum found in part (a) is a global one.

14. A missile has a guidance device which is sensitive to both temperature, t°C, and humidity, h. The range in km over which the missile can be controlled is given by

$$\text{Range} = 27{,}800 - 5t^2 - 6ht - 3h^2 + 400t + 300h.$$

What are the optimal atmospheric conditions for controlling the missile?

15. A closed rectangular box has volume $32\,\text{cm}^3$. What are the lengths of the edges giving the minimum surface area?

16. A closed rectangular box with faces parallel to the coordinate planes has one bottom corner at the origin and the opposite top corner in the first octant on the plane $3x + 2y + z = 1$. What is the maximum volume of such a box?

17. An international airline has a regulation that each passenger can carry a suitcase having the sum of its width, length and height less than or equal to 135 cm. Find the dimensions of the suitcase of maximum volume that a passenger may carry under this regulation.

18. Design a rectangular milk carton box of width w, length l, and height h which holds $512\,\text{cm}^3$ of milk. The sides of the box cost 1 cent/cm^2 and the top and bottom cost 2 cent/cm^2. Find the dimensions of the box that minimize the total cost of materials used.

19. Find the point on the plane $3x+2y+z = 1$ that is closest to the origin by minimizing the square of the distance.

20. What is the shortest distance from the surface $xy + 3x + z^2 = 9$ to the origin?

21. For constants a, b, and c, let $f(x, y) = ax + by + c$ be a linear function, and let R be a region in the xy-plane.
(a) If R is any disk, show that the maximum and minimum values of f on R occur on the boundary of the disk.
(b) If R is any rectangle, show that the maximum and minimum values of f on R occur at the corners of the rectangle. They may occur at other points of the rectangle as well.
(c) Use a graph of the plane $z = f(x, y)$ to explain your answers in parts (a) and (b).

22. Two products are manufactured in quantities q_1 and q_2 and sold at prices of p_1 and p_2, respectively. The cost of producing them is given by

$$C = 2q_1^2 + 2q_2^2 + 10.$$

(a) Find the maximum profit that can be made, assuming the prices are fixed.
(b) Find the rate of change of that maximum profit as p_1 increases.

23. A company operates two plants which manufacture the same item and whose total cost functions are

$$C_1 = 8.5 + 0.03q_1^2 \quad \text{and} \quad C_2 = 5.2 + 0.04q_2^2,$$

where q_1 and q_2 are the quantities produced by each plant. The company is a monopoly. The total quantity demanded, $q = q_1 + q_2$, is related to the price, p, by

$$p = 60 - 0.04q.$$

How much should each plant produce in order to maximize the company's profit?[6]

24. The quantity of a product demanded by consumers is a function of its price. The quantity of one product demanded may also depend on the price of other products. For example, the demand for tea is affected by the price of coffee; the demand for cars is affected by the price of gas. The quantities demanded, q_1 and q_2, of two products depend on their prices, p_1 and p_2, as follows:

$$q_1 = 150 - 2p_1 - p_2$$

$$q_2 = 200 - p_1 - 3p_2.$$

(a) What does the fact that the coefficients of p_1 and p_2 are negative tell you? Give an example of two products that might be related this way.
(b) If one manufacturer sells both products, how should the prices be set to generate the maximum possible revenue? What is that maximum possible revenue?

[6] Adapted from M. Rosser, *Basic Mathematics for Economists*, p. 318 (New York: Routledge, 1993).

25. A company manufactures a product which requires capital and labor to produce. The quantity, Q, of the product manufactured is given by the Cobb-Douglas function

$$Q = AK^a L^b,$$

where K is the quantity of capital; L is the quantity of labor used; and A, a, and b are positive constants with $0 < a < 1$ and $0 < b < 1$. One unit of capital costs $\$k$ and one unit of labor costs $\$\ell$. The price of the product is fixed at $\$p$ per unit.

(a) If $a + b < 1$, how much capital and labor should the company use to maximize its profit?
(b) Is there a maximum profit in the case $a + b = 1$? What about $a + b \geq 1$? Explain.

26. Let $f(x, y) = x^2(y + 1)^3 + y^2$. Show that f has only one critical point, namely $(0, 0)$, and that point is a local minimum but not a global minimum. Contrast this with the case of a function with a single local minimum in one-variable calculus.

27. Find the parabola of the form $y = ax^2 + b$ which best fits the points $(1, 0)$, $(2, 2)$, $(3, 4)$ by minimizing the sum of squares, S, given by

$$S = (a + b)^2 + (4a + b - 2)^2 + (9a + b - 4)^2.$$

28. Find the least squares line for the data points $(0, 4)$, $(1, 3)$, $(2, 1)$.

29. Let $f(x, y) = 80/(xy) + 20y + 10x + 10xy$ in the region R where $x, y > 0$.

(a) Explain why $f(x, y) > f(2, 1)$ at every point in R where
(i) $x > 20$ (ii) $y > 20$
(iii) $x < 0.01$ and $y \leq 20$
(iv) $y < 0.01$ and $x \leq 20$
(b) Explain why f must have a global minimum at a critical point in R.
(c) Explain why f must have a global minimum in R at the point $(2, 1)$.

30. Let $f(x, y) = 2/x + 3/y + 4x + 5y$ in the region R where $x, y > 0$.

(a) Explain why f must have a global minimum at some point in R.
(b) Find the global minimum.

31. **(a)** The energy, E, required to compress a gas from a fixed initial pressure P_0 to a fixed final pressure P_F through an intermediate pressure p is[7]

$$E = \left(\frac{p}{P_0}\right)^2 + \left(\frac{P_F}{p}\right)^2 - 1.$$

How should p be chosen to minimize the energy?
(b) Now suppose the compression takes place in two stages with two intermediate pressures, p_1 and p_2. What choices of p_1 and p_2 minimize the energy if

$$E = \left(\frac{p_1}{P_0}\right)^2 + \left(\frac{p_2}{p_1}\right)^2 + \left(\frac{P_F}{p_2}\right)^2 - 2?$$

32. The Dorfman-Steiner rule shows how a company which has a monopoly should set the price, p, of its product and how much advertising, a, it should buy. The price of advertising is p_a per unit. The quantity, q, of the product sold is given by $q = Kp^{-E}a^{\theta}$, where $K > 0$, $E > 1$, and $0 < \theta < 1$ are constants. The cost to the company to make each item is c.

(a) How does the quantity sold, q, change if the price, p, increases? If the quantity of advertising, a, increases?
(b) Show that the partial derivatives can be written in the form $\partial q/\partial p = -Eq/p$ and $\partial q/\partial a = \theta q/a$.
(c) Explain why profit, π, is given by $\pi = pq - cq - p_a a$.
(d) If the company wants to maximize profit, what must be true of the partial derivatives, $\partial\pi/\partial p$ and $\partial\pi/\partial a$?
(e) Find $\partial\pi/\partial p$ and $\partial\pi/\partial a$.
(f) Use your answers to parts (d) and (e) to show that at maximum profit,

$$\frac{p - c}{p} = \frac{1}{E} \quad \text{and} \quad \frac{p - c}{p_a} = \frac{a}{\theta q}.$$

(g) By dividing your answers in part (f), show that at maximum profit,

$$\frac{p_a a}{pq} = \frac{\theta}{E}.$$

This is the Dorfman-Steiner rule, that the ratio of the advertising budget to revenue does not depend on the price of advertising.

Strengthen Your Understanding

In Problems 33–35, explain what is wrong with the statement.

33. A function having no critical points in a region R cannot have a global maximum in the region.

34. No continuous function has a global minimum on an unbounded region R.

35. If $f(x, y)$ has a local maximum value of 1 at the origin, then the global maximum is 1.

In Problems 36–37, give an example of:

36. A continuous function $f(x, y)$ that has no global maximum and no global minimum on the xy-plane.

37. A function $f(x, y)$ and a region R such that the maximum value of f on R is on the boundary of R.

[7] Adapted from Aris Rutherford, *Discrete Dynamic Programming*, p. 35 (New York: Blaisdell, 1964).

Are the statements in Problems 38–46 true or false? Give reasons for your answer.

38. If P_0 is a global maximum of f, where f is defined on all of 2-space, then P_0 is also a local maximum of f.

39. Every function has a global maximum.

40. The region consisting of all points (x, y) satisfying $x^2 + y^2 < 1$ is bounded.

41. The region consisting of all points (x, y) satisfying $x^2 + y^2 < 1$ is closed.

42. The function $f(x, y) = x^2 + y^2$ has a global minimum on the region $x^2 + y^2 < 1$.

43. The function $f(x, y) = x^2 + y^2$ has a global maximum on the region $x^2 + y^2 < 1$.

44. If P and Q are two distinct points in 2-space, and f has a global maximum at P, then f cannot have a global maximum at Q.

45. The function $f(x, y) = \sin(1 + e^{xy})$ must have a global minimum in the square region $0 \le x \le 1, 0 \le y \le 1$.

46. If P_0 is a global minimum of f on a closed and bounded region, then P_0 need not be a critical point of f.

15.3 CONSTRAINED OPTIMIZATION: LAGRANGE MULTIPLIERS

Many, perhaps most, real optimization problems are constrained by external circumstances. For example, a city wanting to build a public transportation system has only a limited number of tax dollars it can spend on the project. In this section, we see how to find an optimum value under such constraints.

In Section 15.2, we saw how to optimize a function $f(x, y)$ on a region R. If the region R is the entire xy-plane, we have *unconstrained optimization*; if the region R is not the entire xy-plane, that is, if x or y is restricted in some way, then we have *constrained optimization*.

Graphical Approach: Maximizing Production Subject to a Budget Constraint

Suppose we want to maximize the production under a budget constraint. Suppose production, f, is a function of two variables, x and y, which are quantities of two raw materials, and that

$$f(x, y) = x^{2/3}y^{1/3}.$$

If x and y are purchased at prices of p_1 and p_2 thousands of dollars per unit, what is the maximum production f that can be obtained with a budget of c thousand dollars?

To maximize f without regard to the budget, we simply increase x and y. However, the budget constraint prevents us from increasing x and y beyond a certain point. Exactly how does the budget constrain us? With prices of p_1 and p_2, the amount spent on x is p_1x and the amount spent on y is p_2y, so we must have

$$g(x, y) = p_1x + p_2y \le c,$$

where $g(x, y)$ is the total cost of the raw materials and c is the budget in thousands of dollars.

Let's look at the case when $p_1 = p_2 = 1$ and $c = 3.78$. Then

$$x + y \le 3.78.$$

Figure 15.26 shows some contours of f and the budget constraint represented by the line $x+y = 3.78$. Any point on or below the line represents a pair of values of x and y that we can afford. A point on the line completely exhausts the budget, while a point below the line represents values of x and y which can be bought without using up the budget. Any point above the line represents a pair of values that we cannot afford.

To maximize f, we find the point which lies on the level curve with the largest possible value of f *and* which lies within the budget. The point must lie on the budget constraint because we should spend all the available money. Unless we are at the point where the budget constraint is tangent to the contour $f = 2$, we can increase f by moving along the line representing the budget constraint in Figure 15.26. For example, if we are on the line to the left of the point of tangency, moving right on the constraint will increase f; if we are on the line to the right of the point of tangency, moving left will increase f. Thus, the maximum value of f on the budget constraint occurs at the point where the budget constraint is tangent to the contour $f = 2$.